

The Doubling Shift: Cyclotomic Cosets as Orbits of a Finite-Order Multiplier Map

Abstract

Doubling an exponent is a dynamical system.

Its orbits are the cyclotomic cosets.

Its clock is the order of 2 modulo $2^n - 1$.

We strip the system down to its reusable core —
multiplication by one finite-order element —
and read APN-ness off as one of its invariants.

Everything is checked in Lean 4 / Mathlib.

Conference paper call — short, slow-read submission.

1 The picture

Fix n . Work modulo $m = 2^n - 1$.

Pick one exponent e .

Double it, again and again:

$$e, \quad 2e, \quad 4e, \quad \dots, \quad 2^{n-1}e \pmod{2^n - 1}.$$

That is a sequence in $\mathbb{Z}/(2^n - 1)$.

It comes back to e .

One exponent. One periodic orbit.

2 The shift

Doubling once moves the whole orbit one step:

$$(e, 2e, 4e, \dots) \xrightarrow{e \mapsto 2e} (2e, 4e, 8e, \dots).$$

That is the *doubling shift* σ .

Its k -th iterate is multiplication by 2^k :

$$\sigma^k(e) = 2^k e.$$

It is the exponent shadow of Frobenius $x \mapsto x^2$:

$$(x^e)^2 = x^{2e}.$$

Algebra of exponents on one side. A shift on the other.

3 The encoding, in Lean

The shift itself:

```
def shift (m : ℕ) : ZMod m → ZMod m := fun e => 2 * e
```

Its k -th iterate is multiply-by- 2^k :

```
theorem shift_iterate (m : ℕ) (e : ZMod m) (k : ℕ) :  
  (shift m)^[k] e = (2 : ZMod m) ^ k * e
```

The iterate rule *is* the doubling rule.

4 Periodicity

The whole engine is one identity:

$$2^n \equiv 1 \pmod{2^n - 1}.$$

So $\sigma^n = \text{id}$.

Every exponent is periodic, and every orbit length divides n .

```
theorem two_pow_eq_one (n : ℕ) :  
  (2 : ZMod (2 ^ n - 1)) ^ n = 1  
  
theorem shift_periodic (n : ℕ) (e : ZMod (2 ^ n - 1)) :  
  (shift (2 ^ n - 1))^[n] e = e  
  
theorem minimalPeriod_dvd (n : ℕ) (e : ZMod (2 ^ n - 1)) :  
  Function.minimalPeriod (shift (2 ^ n - 1)) e | n
```

Orbits are the 2-cyclotomic cosets mod $2^n - 1$.

5 The number-theory pearl

How fast does the clock tick?

The order of 2 modulo $2^n - 1$ is **exactly** n .

And the same is true for *any* base $b \geq 2$:

$$\text{ord}_{b^n-1}(b) = n.$$

The reason is one line:

$$b^k - 1 < b^n - 1 \quad \text{for } 0 < k < n,$$

so $b^k \not\equiv 1$ before step n .

```
theorem orderOf_base (b n : ℕ) (hb : 2 ≤ b) (hn : 1 ≤ n) :  
  orderOf (b : ZMod (b ^ n - 1)) = n  
  
theorem orderOf_two (n : ℕ) (hn : 1 ≤ n) :  
  orderOf (2 : ZMod (2 ^ n - 1)) = n
```

A crisp repunit fact: $\text{ord}_{b^n-1}(b) = n$, base b general.

6 The principal orbit

Start at the generator $e = 1$.

Its orbit is

$$\{1, 2, 4, \dots, 2^{n-1}\}.$$

The clock pearl says: this orbit has **full** length n .

```
theorem shift_minimalPeriod_one (n : ℕ) (hn : 1 ≤ n) :  
  Function.minimalPeriod (shift (2 ^ n - 1)) 1 = n
```

It is the principal 2-cyclotomic coset.

The longest possible orbit, pinned exactly to n .

7 The shift is invertible

Since $\sigma^n = \text{id}$, the inverse of σ is just σ^{n-1} .

So σ is a permutation of phase space: an automorphism of $\mathbb{Z}/(2^n - 1)$.

```
theorem shift_bijective (n : ℕ) (hn : 1 ≤ n) :  
  Function.Bijective (shift (2 ^ n - 1))
```

Finite order $\Rightarrow$ invertible. An automorphism of the phase space.

8 Orbits *are* cyclotomic cosets

Say e and d are on the same orbit if one reaches the other:

$$\text{SameOrbit} : \quad \exists k, \sigma^k(d) = e.$$

This is an equivalence relation (reflexive, symmetric via periodicity, transitive).

Spelled out on the integers, it is the cyclotomic congruence:

$$e \equiv d \cdot 2^j \pmod{2^n - 1}.$$

```
theorem sameOrbit_iff_modEq (n d e : ℕ) :  
  SameOrbit (2 ^ n - 1) (d : ZMod (2 ^ n - 1))  
    (e : ZMod (2 ^ n - 1)) ↔ ∃  
    j : ℕ, e ≡ d * 2 ^ j [MOD 2 ^ n - 1]
```

Orbits partition exponent space. They are exactly the cosets.

9 The reusable core

None of this is really about 2, or about fields.

Take any monoid and any element c . Let $\sigma_c(x) = c \cdot x$.

Then:

- $\sigma_c^k(x) = c^k x$;
- if $c^d = 1$, then σ_c is periodic of period d and a bijection with inverse σ_c^{d-1} ;
- the orbit relation is an equivalence relation.

```
theorem mulShift_iterate {M} [Monoid M] (c : M) (e : M) (k : ℕ) :  
  (mulShift c)^[k] e = c ^ k * e  
  
theorem mulShift_bijective {M} [Monoid M] {c : M} {d : ℕ}  
  (hd : 1 ≤ d) (hc : c ^ d = 1) :  
  Function.Bijective (mulShift c)
```

Multiplication by a finite-order element is a finite-order permutation. The doubling shift is just the case $c = 2$.

10 The Frobenius conjugacy

Why 2? Because of Frobenius.

In any commutative monoid:

$$(x^e)^2 = x^{2e}.$$

Doubling the exponent is post-composing with $x \mapsto x^2$.

So the power map commutes with Frobenius:

$$(x^e)^2 = (x^2)^e.$$

```
theorem frobenius_conjugates_pow {M} [CommMonoid M]
  (e : ℕ) (x : M) : (x ^ e) ^ 2 = x ^ (2 * e)

theorem pow_comm_frobenius {M} [CommMonoid M] (e : ℕ) :
  (fun x : M => (x ^ e) ^ 2) = (fun x : M => (x ^ 2) ^ e)
```

The doubling shift is the exponent-space shadow of Frobenius.

11 APN, as a conjugacy invariant

Here is the payoff slogan.

Any predicate on exponents that is *constant on $\langle 2 \rangle$ -orbits* descends to a predicate on the orbit space.

APN-ness of the power map $x \mapsto x^d$ on $\mathbb{F}_{2^n}$ is such a predicate:

$$d \sim e \quad (\text{same orbit}) \implies (x^d \text{ APN} \iff x^e \text{ APN}).$$

So APN-ness is a **conjugacy invariant of the doubling dynamics**: one orbit, one verdict.

“Same orbit, same verdict.” The dynamical reading of cyclotomic invariance. The field-level proof of this step lives in the project’s APN layer; here we isolate the dynamics it rests on.

12 What we claim

- One exponent $\mapsto$ one periodic orbit under $e \mapsto 2e$.
- $\sigma^k =$ multiply-by- 2^k ; $\sigma^n = \text{id}$.
- Orbit lengths divide n ; orbits are the cyclotomic cosets.
- $\text{ord}_{b^n-1}(b) = n$ for every base $b \geq 2$.
- The generator 1 has full minimal period n .
- σ is a bijection (inverse σ^{n-1}).
- The core is general: $x \mapsto cx$ for a finite-order c .
- APN-ness is a conjugacy invariant: same orbit, same verdict.
- All boxed statements are machine-checked, sorry-free.

Read slowly. Doubling was a dynamical system all along.